LINEAR SYSTEMS · BY SHAPE

What the shape of A predicts

Before any reduction, the shape of A narrows what can happen. It never settles the question — $\mathbf{b}$ still decides existence — but it says which outcomes are available and which are ruled out.

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SHAPES

TYPE	CONDITION	EXISTENCE	UNIQUENESS	PRACTICAL APPROACH
SQUARE				
 **Nonsingular**	$m = n$, $\det A \neq 0$	always, for every $\mathbf{b}$	always unique	factorize once, solve per $\mathbf{b}$
 **Singular**	$m = n$, $\det A = 0$	iff $\mathbf{b} \in \mathrm{Col}(A)$	never, when it exists	reduce $[A \mid \mathbf{b}]$, compare ran
MORE EQUATIONS THAN UNKNOWNS				
 **Overdetermined**	$m > n$ — tall	unusual — most $\mathbf{b}$ miss $\mathrm{Col}(A)$	unique when it exists and $\mathrm{rank}(A) = n$	least squares when inconsistent
MORE UNKNOWNS THAN EQUATIONS				
 **Underdetermined**	$m < n$ — wide	iff $\mathrm{rank}(A) = \mathrm{rank}([A \mid \mathbf{b}])$	never — at least $n - m$ free parameters	impose a criterion: minimum norm, sparsity

Shape constrains, rank decides. A wide matrix cannot have a unique solution whatever $\mathbf{b}$ is; a tall one usually has none. What no shape determines is existence, which depends on whether $\mathbf{b}$ lies in the column space — and that is a fact about $\mathbf{b}$, not about the dimensions.